

Exploration of Rural Patient Critical Decision Making in Emergency Medical Situations

SYDE 642: Cognitive Engineering Methods

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Abstract

Background: Inequities remain when it comes to accessing medical care in rural communities in Canada. In emergency health situations, this population is often required to factor in additional logistics, such as time and transportation, in addition to managing the health concern at hand. The decision not to seek urgent care can have an overall impact on future health outcomes and overall well-being. This study investigated the decision-making processes behind accessing urgent medical care rurally and patient perspectives on reducing barriers to accessing emergency rural health care in general.

Method: This study used the critical decision-making framework to guide interviews to understand patient perspectives on the decision to access rural emergency health care. A focus group session with rural patients was held to discuss potential solutions to barriers.

Results: Decision-making for rural health emergencies may require assessment of various factors, including transportation time and means, wait times, obligations left unattended and what home remedies are available that could be utilized first. Severity of the injury was the most critical factor that was assessed first and was the determining factor in the decision to access care directly or take time to consider other factors involved.

Implications: Various factors impact decision-making when located rurally during medical emergencies. Although larger healthcare system changes would be ideal, focus on other factors and burdens placed on rural residents that affect their ability to decide if and when to access care could be a target in reducing barriers to access to care.

Introduction

Accessible critical healthcare services in rural regions represent a significant challenge in human factors implications. Compared to urban populations, rural residents face higher barriers in accessing quality care such as geographical isolation, limited provider availability, transportation difficulties, and limited access to specialized medical resources (Haggerty et al., 2014). The Canadian healthcare system has implemented initiatives to address these widespread disparities between urban and rural regions (Wilson et al., 2020). However, there remains a disproportionate amount of research focused on urban settings, leaving critical gaps in knowledge relating to understanding and optimizing the rural healthcare system for rural populations.

Existing literature on healthcare access in rural locations has made important contributions to understanding critical decision making made by key players in the healthcare system. While top-down approaches may focus on perspectives of providers, policymakers, and system administrators, research has shown that the inclusion of bottom-up approaches involving patients in healthcare decision-making is associated with improved healthcare outcomes (Vahdat et al., 2014). Numerous studies have explored healthcare decision-making across diverse populations, ranging from investigating barriers faced by patients accessing healthcare in rural British Columbia to analyzing the influence of social pressures in medical decisions faced patients from a rural South Indian hospital (Kornelsen et al., 2021; Seetharam & Zanotti, 2009). However, due

to the large range of rural regions, it is clear that these rural studies are often extrapolated to other rural regions to account for the lack of research done. For example, many central states of America lack their own centralized rural research on healthcare access and rely on research by adjacent states (Douthit et al., 2015). Consequently, it is of great importance to contribute to the growing body of literature on rural research to minimize assumptions and extrapolations in healthcare decision making.

Current research in decision making for accessing healthcare in rural regions of Canada at the patient level employs a focus-group heavy approach with consideration of other methods. Kornelsen et al. (2021) conducted a study of rural British Columbia residents on accessing healthcare (n = 1158) and revealed a need for a framework addressing the concerns of population desires while maintaining the realistic capabilities of healthcare systems (Kornelsen et al., 2021). Wong & Regan (2009) conducted a focus-group study (n = 50) investigating the tradeoffs made by populations living rurally in British Columbia to highlight inequities such as out-of-pocket costs like transportation that reduce healthcare accessibility (Wong & Regan, 2009). Burnett et al. (2020) also implemented a focus-group (n = 72) and interview-based (n = 14) study involving indigenous populations of Northern Ontario that illustrated location and relationships with healthcare providers/states were major factors in decisions to access healthcare (Burnett et al., 2020). Haggerty et al. (2014) used the focus group method (n = 96) to investigate relatively adaptable methods that rural patients received healthcare (telemedicine, home visits, flexible appointments, etc.) in comparison to more rigid methods in their rural counterparts (Haggerty et al., 2014). While there is a growing body of literature on healthcare access and decision-making focused in Canada, these studies lack an overarching framework that can integrate these findings. To our knowledge, structured decision-making models, such as decision-making ladders, have not been applied in this context.

A popular human factors engineering method, the Critical Decision Method (CDM), is a method based in Cognitive Task Analysis that focuses on the utilization of semi-structured interviews to gain a better understanding of decision-making among experts (Jenkins et al., 2013). It has applications in a wide variety of fields, including healthcare (Jenkins et al., 2013). Although traditionally experts may be viewed as healthcare workers in this domain, with regards to decisions surrounding care access, it can be argued that the patients who are seeking care are the primary expert decision-makers in this context (Kennedy, 2003). Consequently, this study aims to address the limitations in understanding rural healthcare access in Ontario by investigating the decision-making process in healthcare access for critical healthcare emergencies from the patient perspective. This study will address the gap in the literature for decision-making frameworks by developing a decision-making ladder that captures the key stages of rural patient decisions when seeking healthcare services.

Methods

Recruitment

Individuals who reside in rural areas of Ontario or who accessed urgent or emergency medical services while located in rural areas of Ontario were eligible for this study. For the sake of this study, eligible rural areas included any location in Ontario outside of a city or town. The medical emergency in question could have occurred at any point in the individual's life. Participants were required to have access to a device that can host Microsoft Teams or a phone call or the ability to meet with the study team in-person.

Study design

Once eligible participants were recruited, they were first invited to participate in an individual CDM interview with our study team. After the completion of all CDM interviews, participants were asked if they could join a focus group session which would involve attending a virtual meeting with other participants and the research team to discuss changes to the rural health system and share whatever perspectives they felt comfortable disclosing to the group and participating in conversation with other attendees. Findings from the CDM interviews and focus group sessions were thematically analyzed and used to develop knowledge representations. An overview of the study design is shown in Appendix 1.

Ethics approval

No research ethics board approval was obtained for this study as it was completed as part of a course project for SYDE 642 - Cognitive Engineering Methods at the University of Waterloo. Participant involvement with this activity was voluntary. No identifiable information was recorded in the study results or in any other way associated with this work, and participants were able to exit the study at any time.

Data collection

Critical Decision Method

For the CDM interview, participants were interviewed either in person, online (using Microsoft Teams) or over the phone. For this study, only one study team member was required to be present in the CDM interview. The CDM protocol used in this study consisted of 9 steps as recommended in Human Factors Methods: A Practical Guide for Engineering and Design (Jenkins et al., 2013). The selected CDM probes are presented in Appendix 2.

Before beginning interviews, a study team member would converse with the participant briefly to build rapport before beginning the series of questions. Then participants were asked to think of a time where they accessed emergency medical care rurally. Participants were asked to describe the series of this event in detail in their account. After this was discussed, participants were then asked more specific questions using the preselected CDM probes (Appendix 2) to better assess the situation. Finally, participants were then asked specific "what-if" or hypothetical situation questions relating to the incident they described. Before finishing the interview, participants were asked demographic questions. All interviews were recorded and transcribed. Interviews were rewatched by a study team member to confirm transcript accuracy.

Focus groups

A virtual focus group session took place over Microsoft Teams. Participants were asked to unmute themselves during this call to answer questions and participate in the discussion but were not required to turn on their camera unless they wished to.

The focus group discussion was guided by five questions developed by the study team prior to the event. The five questions were:

- 1) What's your experience been like accessing healthcare in a rural area?
- 2) What factors influence your decision to seek care?
- 3) What challenges do you face when trying to access healthcare?
- 4) What is a specific example/experience you have with accessing rural healthcare
- 5) What one change would make rural healthcare access better for you?

Participants were asked to share their responses to each question. Responses were optional, and participants could skip responding. Participants could also respond in full or in part to another statement or response by another attendee. The focus group session was recorded and transcribed using Microsoft Teams.

Data Analysis

For the thematic analysis of both the CDM interviews and the focus group sessions, interview transcripts were first inductively coded by one study team member using the comment feature on Microsoft Word. This code was then reviewed by an additional team member. Any conflicts were discussed with a third member of the study team. Thematic analysis was then conducted by either one or two study team members. The findings from each thematic analysis were reviewed by all three study team members. Proportions were presented for participant characteristics for identified gender and occupation. Descriptive statistics (mean) of overall participant age was determined. Additionally, a word-frequency analysis was conducted on Microsoft Word using the Find tool. Instances of words selected were counted for each interview done.

Knowledge representation development

Upon completion of the thematic analysis, these findings guided the development of different forms of knowledge representations. For CDM, CDM finding tables were developed as a result of the CDM interview process. The findings from the CDM tables, in addition to the thematic analysis from the focus group session guided the creation of decision ladders.

Results and Analysis

Population

This study conducted a series of semi-structured interviews adapted for contextual inquiry purposes over a timeframe of two weeks. Three adult participants (n = 3) participated in these

interviews and were later included in a focus group. Demographic information on these participants is provided in the table below.

Table 1. Participant characteristics

Demographics	
Mean Age	26.67 (n = 3)
Percent Male	66.67% (2)
Percent Female	33.33% (1)
Occupations	Nursing Student (P1) Quantum NeuroPharm PhD Student (P2) Medical Student (P3)
Locations of Injury	Forester Falls, Redford County, ON. Conestoga Lake, ON. Port Sydney, ON.

CDM findings

For each interview, a timeline of the critical timeframe leading up to, including and following the time of the injury was created for each participant. A sample timeline from this study is included in Appendix 3. The CDM table resulting from the scenario phases is provided below in Appendix 4 (Jenkins et al., 2013). Themes emerging from the CDM interviews are described below in the following section. A short quantitative analysis on frequency of words is provided in Appendix item 6.

Thematic analysis from CDM Interview

Personal and Perceptual Factors Influencing Healthcare-Seeking Behavior

Several related and interconnected themes emerged from the interviews on perceptions of seeking healthcare and the personal factors that influenced such behavior. Many of these perceptions were negative and two main subthemes were revealed after coding: (1) Avoidance of healthcare and (2) Delayed seeking healthcare. It appears that participants often delayed seeking healthcare, specifically in rural regions, because of potential long wait-times. In fact, several participants mentioned waiting to see if the injury would worsen before seeking medical professional help. For instance, one participant waited several hours to see if their injury would stop bleeding, and another waited an entire night before seeking medical care.

“Even after, several hours of constant pressure applied to the wound and using tape to try and hold it closed, it continued to bleed.” (P1)

“Yeah, something might be wrong [reference to injury]. It [didn’t] look that bad to me. It’s probably just a bruise. Like, [I’ll] reassess in the morning. So, I slept. Not very well. I slept sitting up. I knew that there was something wrong.” (P3)

Consequently, when the participants were asked if anything would have changed if they had been in an urban region, they mentioned that it would be significantly easier to seek medical care. None of the participants mention issues of wait-times for urban regions.

“Oh, then it’s easy to just go to a walk-in clinic or two [referring to urban emergency healthcare]. Like a basic emerge or something. Then I would definitely [go]. [If] I was living in the city, or something, I definitely would go.” (P1)

Structural and Systemic Barriers to Accessing Healthcare

There were several barriers to accessing healthcare in rural areas. One of the main structural barriers included transportation as none of the participants used an ambulance for accessing emergency medical care. Additionally, participants were hesitant to use public transport in a hypothetical situation, which left personal transportation as the only solution. Two of the participants had to rely on family members for transportation to hospitals and the third participant had to drive themselves. While the cost of transportation was not extensive because all participants were within thirty minutes of the nearest hospital, this reveals a troubling trend that rural healthcare is only accessible if one has a personal mode of transportation. The concept of a “walk-in” clinic is not an available option to the majority of those that live in rural areas.

Additionally, a systematic barrier that further compounds this issue is the fact that many of the patients mentioned that receiving medical care from specialists or specific equipment required further transportation to an urban center.

“[That’s] an unfortunate consequence of having less people in [rural] areas, right, because [the government] is not going to spend, you know, a couple million dollars on an MRI machine in a town of only, like 18,000 people. You kind of understand that’s the problem and there’s no easy solution, unfortunately.” (P3)

Additionally, another systemic barrier that participants mentioned was wait-times. All participants mentioned waiting for medical care from as little as 30 minutes to six hours just to see the medical practitioner. Additionally, the entire process from triage to exit took as long as 12 hours for some participants. Long wait-times are a systemic issue of the Canadian healthcare system, but they are often exacerbated in rural regions (Haggerty et al., 2014). When asked what they (the participants) expected from seeking healthcare, one participant answered succinctly:

“That I would spend the rest of my day there.” (P1)

Internal Knowledge, Judgment, and Self-Reliance in Healthcare Decision-Making

Prior knowledge heavily influenced the decision-making process of participants in seeking healthcare. Two of the participants have a medical background and had experienced the healthcare system in rural areas before. This seemed to provide a basis for their decisions making: (1) One participant delayed going to healthcare for as long as they could, deciding to rely on their own medical knowledge to try and fix the injury, (2) one participant relied on familial pressure to seek medical treatment, and (3) one participant relied on external medical professional opinion

(paramedic) before seeking treatment. In all three cases, the participants referenced reliance on their own medical knowledge and experiences to try and assuage the situation from getting worse. None of them immediately decided to seek professional medical care.

The degree to which each participant's prior medical knowledge helped them seemed apparent with P1 considering drastic measures to prevent going to the hospital while P2 and P3 did not consider alternatives (due to no current or limited medical background).

"Taking some of [the large animal] freezing medication and freezing my finger and having [my finger] stitched up. To avoid having to go to town and the long wait and things. Yeah, that was a consideration." (P1)

External Factors and Symptom Severity Driving Healthcare Seeking

Several external factors ultimately drove participants to seek healthcare. One such major factor was familial pressure. Two participants cited their family pressuring them to seek medical care and another two cited family helping transport them to the hospital. The availability of family support can influence a person's decision to seek healthcare and ultimately, to advocate for healthcare services too. In one case, a participant's family also influenced the decisions of the medical practitioner.

"[The practitioner] was actually reluctant to set the broken bridge of my nose back into place. At that point, my mom didn't like that idea. To leave it in place, leave it as it is, and eventually convinced the doctor to force my nose bridge back into place using his hands." (P2)

Other external factors included symptom severity with many participants citing the amount of blood from the injury as a main factor for seeking healthcare. Additionally, pain was another factor with one participant stating that they sought medical care because they wanted to reduce the pain they were in.

Perceptions and Beliefs About the Healthcare System

When asked about their perceptions or beliefs of the Canadian healthcare system, expectations and reality played a role in shaping participant's judgements. One participant expressed negative beliefs about the healthcare system stating that practitioners displayed low-empathy and minimal compassion. This perception was continuously reinforced by prior negative experiences with medical practitioners.

"It might have to do with the fact that the average person [the practitioner] sees probably goes to the hospital or the clinic at the drop of a hat. However, I feel that since I am of a subset that doesn't like to go to the doctor for no damn reason, when I get there, they're primed to dismiss my complaints." (P2)

Conversely, another participant had a positive view of the healthcare system due to prior positive experiences. They expressed their gratitude for rural healthcare access near the location where they had been injured and stated trust in the system that trained the practitioners. Despite this, they too mentioned having a disappointing healthcare experience.

“And I remember being disappointed that they weren’t giving me any more pain medications or something like that because that’s what they would normally do for a broken bone other than Advil. But I was also, you know, 15 or 16. I didn’t really know how the healthcare system worked at that point.” (P3)

It seems that expectations of the current healthcare system are influential in reinforcing perceptions of the healthcare system. In the cases above, when expectations were not met, negative perceptions of the system were formed (potentially reinforcing these perceptions). However, matching expectations with reality could potentially shift and mitigate the formation of negative perceptions on the healthcare system. Additionally, transparent decision-making on the practitioner's side can potentially help enable patients to understand why they are receiving the treatment they receive. As mentioned by Kornelsen et al. (2021) in matching patient expectations to system needs at a high-level, matching patient expectation of healthcare received with the reality of actual healthcare available by practitioners is integral to ensuring quality healthcare (Kornelsen et al., 2021).

Additionally, two of the participants commented on the state of the current healthcare system and issues surrounding logistics and resource management. They both mentioned a lack of resources for rural healthcare systems and cited this as one of the main reasons for many of the disadvantages surrounding rural healthcare.

“[The rural healthcare system] is thinly stretched in the sense of where coverage is very minimal. So, staff are overworked and under supported and so that’s why things can take longer.” (P1)

Thematic analysis from the focus group

The focus group discussion led to an interactive conversation that revealed responses and themes in four main areas: experiences overall with rural healthcare, barriers to accessing care, logistics involved in care access decisions and changes proposed for the rural health system. The overarching goal of the focus group discussion was to better understand the overall barriers impacting access to care and what potential changes could be implemented to mitigate such barriers.

Experiences with the healthcare system

When asked what their overall experience was accessing rural healthcare, participants often chose not to access care or would wait to see if symptoms would lessen before going to access care. When they did access care, there was some variation regarding experience with the quality of care in rural centers, as one participant said it was comparable in quality to care received in urban settings, whereas another said they felt that they needed to have an external party advocate for them to get the care they needed.

“Usually the care is, you know, maybe still pretty good quality compared to what you get in the city, but it just takes so much longer to access it with wait times and lack of resources and all of that.” (P3)

*“And he didn't even wanna fix it.
He wasn't going to set the broken bone or anything, and it was only after, like, my mom,
specifically telling him that she wants it fixed, that he like pushed my broken nose back into
place.” (P2)*

Another experience evident from this discussion was the oftentimes preference to manage conditions at home with whatever resources were available, even if these resources were not utilized for their intended purpose, as opposed to going in and accessing emergency medical care.

“I'm also fortunate that my [parent] is actually a large animal veterinarian, so although not necessarily supposed to, we do have some access to pharmaceuticals that not the general population does.

So it's kind of like if it can be managed at home and not have to go and spend an entire day in hospital. [Then it does.] Then we do.” (P1)

Barriers to accessing care

Barriers were brought up in this conversation both when directly asked as well as when discussing their overall experience with rural health, indicating this is an important aspect of the experience of interacting with the healthcare system. Main barriers discussed included transportation, such as needing to drive.

“Most of the time I choose not to go into um, say a clinic or hospital, simply because uh or I lean towards not going into clinic or hospital just because of like [some] barriers sometimes due to like driving and whatnot.” (P2)

Another common barrier mentioned in this discussion was the inability to take time and leave their house due to obligations or responsibilities at home or on the farm. In addition, knowing that there was likely going to be a long wait time when they arrived at the hospital was also listed as a barrier, or deterrent, to accessing care, especially if it was felt the condition could be managed at home.

*“So like, that's [a] big, like, my experience is like I really don't go very often because.
[If] I can just manage it at home, than I do.” (P1)*

Logistics involved in care access

Patients in this discussion were aware that, often times there would only be one emergency doctor present in the emergency room at the rural hospitals, which played into their decision-making. It was also mentioned that a lack of other health providers in the ER was also likely and knowledge of this was used in their decision-making. These two factors tied into the main logistical factor discussed, which was that patients knew they would be sacrificing a lot of time if they did choose to access care. That included time driving to the health center in addition to time spent waiting, due to limited resources.

“So it's kind of like if you're going to go in for something, it needs to be something serious because you're going to spend your entire day there.” (P1)

In addition to time spent, another logistic involved figuring out the means of transportation. Some views in the discussion felt more willing to utilize paramedic services rurally, while others claimed that was often too extreme.

“If you can't drive yourself, then someone else has to drive you, or you have to make the leap to call like 911 and have an ambulance come and get you. And that's just, that's a big thing really. And I think a lot of people in rural areas just don't want to make a big thing out of it.” (P2)

Although not a direct experience among the group, it was highlighted that for even more remote rural residents, logistical decision making likely would need to include accommodation and flights to accessing care.

“So I think accommodation and transportation are two of the big ones. 'cause, if you're flying to get healthcare, you're probably gonna have to stay for a couple days and usually you're not able to stay in the hospital unless it's super critical. So, if you even have a hospital bed, right?” (P3)

Proposed changes

Participants wished to see a larger system change when asked about changes to rural health, although it was then said they know that is likely not feasible. In the meantime, some other changes could be changes to the medical curriculum and resident recruitment efforts to improve the appeal of rural medicine, resulting in more physicians. This would help address the other recommendation of reducing rural physician workload and improving understanding of rural patient needs among rural care providers.

“So I think maybe we would have better, better health care outcomes if these rural doctors weren't so overworked. And that could change.” (P2)

“Programs mostly residency programs, making family medicine more enticing.” (P3)

Decision ladder

Qualitative data collected in the two phases of this study was used to develop a decision ladder which shows the patient decision-making process undertaken when deciding if to access healthcare following an injury (Appendix 5).

Conclusions and Limitations

In this study, our research team explored perspectives on decision-making for urgent health access among rural patients. The themes that emerged from this study indicate that the severity of injury was a highly important factor that was considered by rural patients in their decision to access care. Findings suggest that if deemed highly serious, likely equated to life-threatening, then individuals would be more likely to proceed directly to executing care access procedures. However, if severity is deemed to be anything less than this, other factors are considered that impact their decision to

either access care at all, or if they will access care, when that will be. Delay in accessing care allowed for attempted self-management using at-home resources, or to see if symptoms worsen over time. This theme was not surprising, as rural communities have been shown to have higher levels of self-reliance when it comes to healthcare (Page-Carruth et al., 2014). However, this trend in self-reliance among these populations may also be a contributing factor to worse health outcomes observed in rural communities (DeGuzman et al., 2022; Santore et al., 2025).

Other factors identified as being involved in the decision-making process in this study included increased amounts of logistical planning, such as means of transportation and time spent away from home and the implications of their absence. Although likely not a unique experience to only rural residents, the impact of these factors may be amplified in this population by the increased distance from hospitals, in addition to the knowledge that there are likely limited resources available at the hospital. Consequently, knowing the likelihood of increased travel and wait times and the likely delay back home to resume their duties, either around the house or farm, may have a unique impact on rural patients and their decision-making. Another related factor involves attitudes towards and trust in rural healthcare workers, which was found to be mixed among participants, although an understanding of extremely high workload was consistently present. A potential explanation for this could be centered around the fact that rural residents place so much effort on their triage and may view, which could result in feelings of mistrust and frustration when it is perceived that healthcare workers may not appreciate the participants experience of overcoming personal biases rooted in self-reliance and preferences for home remedies, in addition to their value in their ability to triage and manage their own health. This likely is not the only contributing factor, as other literature also found that mistrust can stem from negative attitudes and a lack of funding allocated towards rural health (Lister & Joudrey, 2023).

This study employed a unique take on the traditional CDM framework, which highlighted the utility of this method for the investigation of non-routine and emergency situations. (Jenkins et al., 2013). This study had some limitations to consider when interpreting these findings. The main limitation of this study is the small sample size. Only three participants were able to be recruited, who met our inclusion criteria, in the timeframe permitted for the current study. The small sample size has resulted in a lack of ability to perform meaningful statistical analysis, indicating that we cannot state any significance of these findings. Furthermore, all of the participants recruited had some form of and awareness of medical knowledge and/or training. This has implications in our findings, as individuals in this sample may be more in tune with thinking about certain aspects of health that those in the general population may be, which can result in non-generalizable findings. Therefore, future work must consider more diversity in educational and professional backgrounds among rural participants.

Another limitation of this study was the definition of rural used. This study allowed rural to be defined as any area in Ontario outside of a city or town. The definition was selected to ensure some participants could be recruited in the permitted timeline of this work, as a more restrictive definition may have impacted recruitment efforts. However, any continuation of this work should consider a more structured definition of rural, as it is not accurate to assume that everyone deemed

to be living outside of a city or town will have the same experience. There are wide difference within rural populations that would be important to capture in this work. A more structured definition would then in turn, allow for the ability to conduct meaningful analysis to detect differences *within* various rural populations.

Some additional considerations about limitations of this study were that participants were allowed to recount any medical emergency that happened at any point in their life. Events that occurred longer ago may be more difficult to remember, which has the potential to impact accuracy. Finally, there was no criterion instilled on the severity of medical emergencies allowed to be discussed. This was decided as to avoid imply that any medical emergency experienced by the participant was deemed more or less “important” or “serious”; however, future work should consider determining a way to categorize the nature of the injuries, as this impacts the decision to access to care or not and should be accounted for in the analysis.

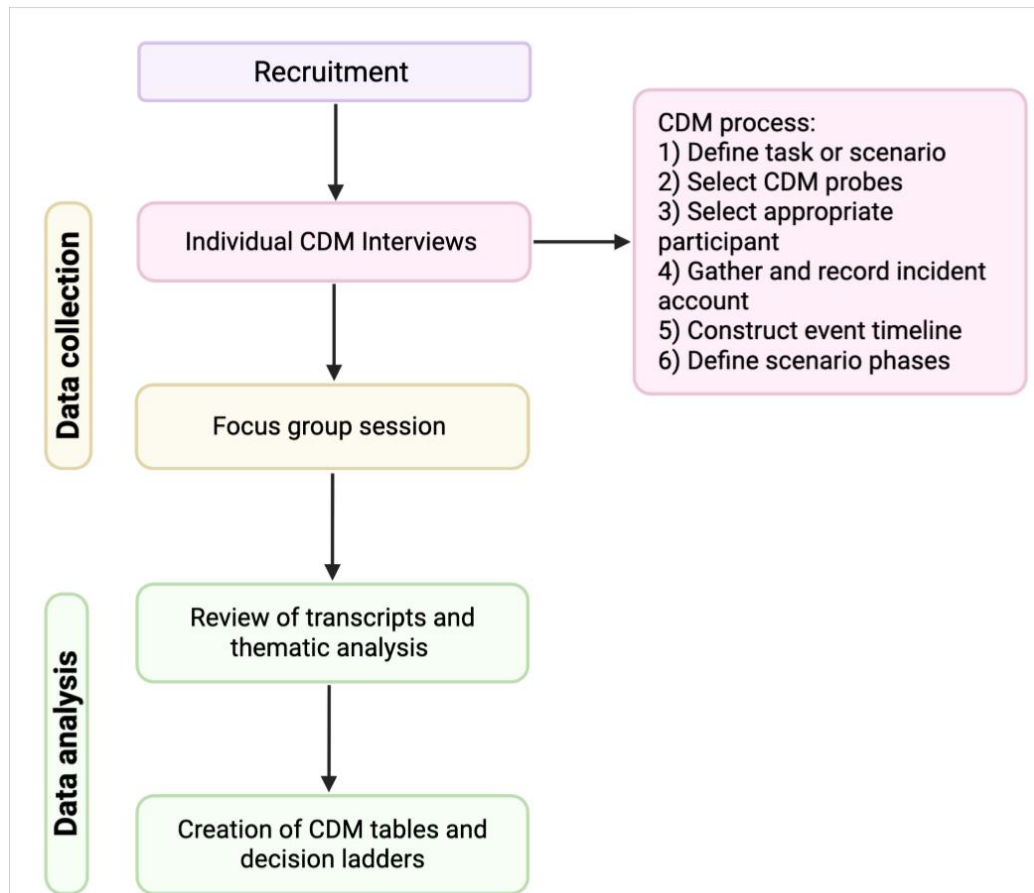
Overall, this study established a unique CDM protocol that can be used to investigate emergency health access decision-making among rural populations. However, a larger study with a more diverse population is needed to better understand these emerging themes and further understand nuanced differences between different rural communities and populations. Findings from this report indicate that to address these known barriers to care, we must consider not only addressing the system issues that result in long wait times and physical barriers (such as distance) inhibiting easy access to care. But further, knowledge translation and public outreach efforts could perhaps be useful in these communities to further explore the identified factors in this study that play a major role in deciding if and when to access care.

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Appendix 1. Study protocol flowchart



Appendix 2. Critical Decision-Making Probes

Probe Area	Probe Question
Cue Identification	What specific signs, symptoms, or circumstances made you realize you needed to seek healthcare?
Expectancy	What did you expect would happen when you decided to seek care in your rural area?
Conceptual	What are your beliefs and understandings of the rural healthcare system?
Influence of Uncertainty	How did not knowing the outcome or the quality of care affect your decision-making process?
Information Integration	How did you weigh the pros and cons of seeking care, such as cost, distance, or severity of your condition?
Situation Awareness	What information did you have to determine how urgent or serious your healthcare need was?
Situation Assessment	What factors made you decide to seek care immediately versus waiting or avoiding it altogether?
Options	What alternatives to formal healthcare (e.g., home remedies, telehealth) did you consider, and why?
Decision Blocking – Stress	What challenges, like financial, emotional, or logistical stress, made it harder to decide to seek care?
Basis of Choice	What was the most important factor that ultimately led you to your final decision?
Analogy/Generalization	Did you rely on past experiences or similar situations to guide your decision, and how did they help?

Appendix 3. Critical Decision Method Timeline of Seeking Rural Healthcare (P1)

Timeline (time spent on event)	Event
Start time.	Injury event occurs.

“several hours later” (exact time not specified)	Patient decides to seek healthcare after alternative solutions fail.
25 minutes	Patient drives to hospital.
15 minutes	Patient passes through triage.
3 – 4 hours	Patient waits to see nurse or doctor.
2 hours	Patient sees doctor. Doctor prescribes x-ray scans. Patient waits for x-ray tech to arrive.
1.5 hours	X-rays scans are complete. Patient waits for result of x-ray scans.
1 hour	X-ray results are read.
1 hour	Final Doctor seen. Stitches are administered. Patient is released from hospital.
25 minutes	Patient drives back home.

Appendix 4. CDM Table: Deciding to Access Rural Healthcare

Goal Specification	To seek timely and effective medical care for an injury/illness in a rural setting.
Cue Identification	Persistent bleeding Severe/unrelenting pain Family pressure
Expectancy	Long wait times Limited local resources Rushed improper care
Conceptual Model	Rural healthcare is under-resourced and overburdened Urban healthcare is more comprehensive Self-reliance is preferred unless symptoms escalate
Uncertainty	Whether injury will worsen without care Uncertainty about wait times and treatment quality Whether transportation is available
Information	Prior medical knowledge Past negative/positive healthcare experiences Advice from family
Situation Awareness	Recognizing that self-treatment is insufficient Realizing injury severity Assessing transportation options
Situation Assessment	Weighing risks of delaying care vs. long wait times Elimination of alternative self-care options Evaluating if urban care is worth the extra travel
Options	Delay care and self-treat Seek rural healthcare despite wait times Travel to urban center if possible
Stress	Pain/discomfort from injury Anxiety about long waits Frustration with systemic barriers
Choice	Ultimately seek rural healthcare due to symptom severity/family pressure
Analogy	Comparing rural vs. urban healthcare access Relating to past experiences in healthcare

Appendix 5. Urgent Rural Health Access Decision Ladder



Appendix 6. Frequency Analysis Table: Main Themes of Common Words Stated During Interviews

Theme	Word	Frequency (P1, P2, P3)	Average
Providers and Settings	doctor, physician, nurse, specialist, clinic, hospital, telehealth, pharmacy, practitioner	13, 39, 27	26.3
Access and Logistics	access, availability, distance, transportation, travel, appointment, waiting, schedule, referral, cost, price	13, 11, 21	15.0
Information and Communication	information, knowledge, advice, educate, inform, communicate, explain	3, 2, 6	3.6
Social Factors	family, trust, belief, support	4, 5, 8	5.6
Decisions	decide, choice, prefer, confident, unsure, concern, experience	12, 16, 5	11.0